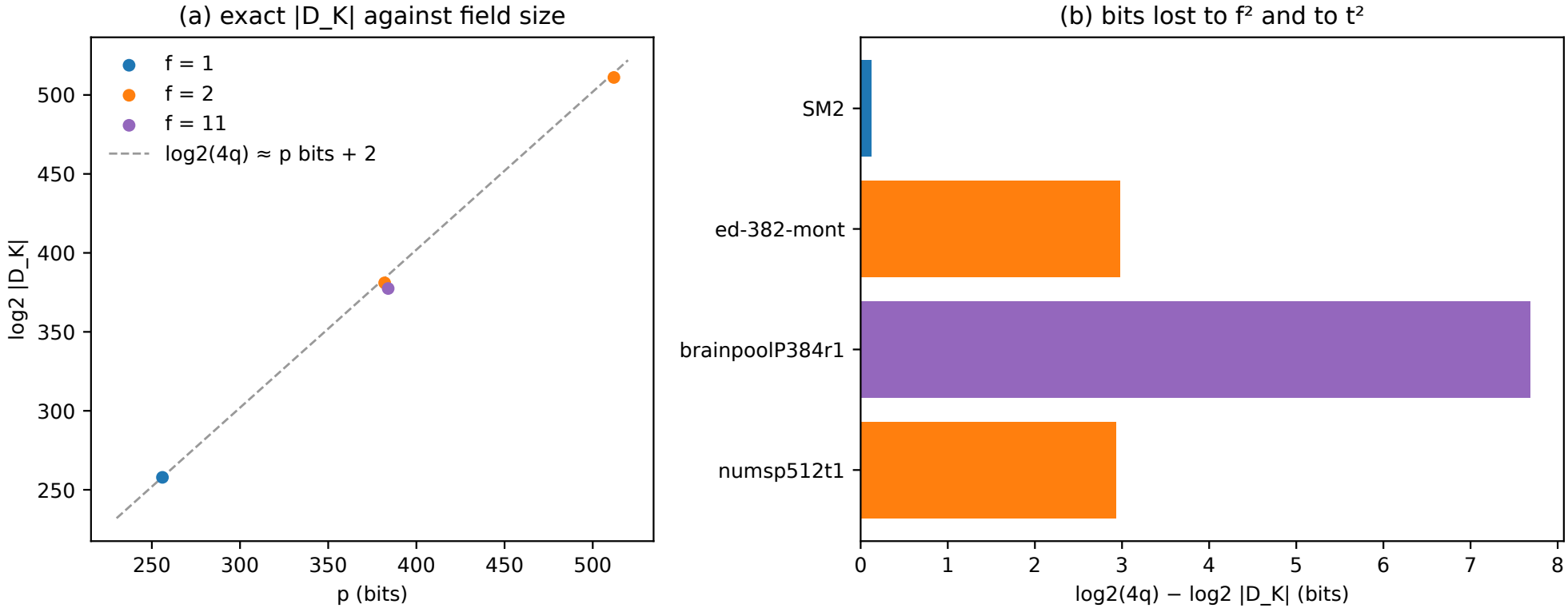
Exact CM discriminants of 6 more standardized curves (batch 2)

research/endosweep_dk_exact_batch2_20261016 (2026-10-16). Data: `dk_exact.json`, `factorizations.jsonl`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration. They are the six curves the factoring driver completed between the frozen snapshot of batch 1 (research/endosweep_dk_exact_20261016) and the second frozen snapshot, five of them by a CADO-NFS split. std-curves records no `cm_disc` for any of the six, so none was checked against the literature.



n = 4 curves; exact integers (PARI isprime-proved factorisations), no uncertainty. Not shown: D_K = −3 on bn/bn606; D_K = −339 on arkworks/cp6_782.

Figure 1: (a) $\log_2|D_K|$ against p for the 4 curves with a large D_K, coloured by the conductor f. (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`. Not drawn: $D_K = -3$ on bn606 and $D_K = -339$ on cp6_782.

Results

curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
oscaa/SM2	256	78	78	1	(78 digits)	257.88	2^255.88	no cm_disc
nums/ed-382-mont	382	116	115	2	(115 digits)	381.02	2^379.02	no cm_disc
brainpool/brainpoolP384r1	384	116	114	11	(114 digits)	377.45	2^375.45	no cm_disc
nums/numsp512t1	512	155	154	2	(154 digits)	511.07	2^509.07	no cm_disc
bn/bn606	606	183	1	(92 digits)	-3	1.58	1	no cm_disc
arkworks/cp6_782	782	236	3	(117 digits)	-339	8.41	85	no cm_disc

Excluded: P-256, P-384, P-521, Curve25519 (research/endosweep_nist_20261006) and the 18 curves of research/endosweep_dk_exact_20261016. Lean: `CryptoResearch.Endosweep.DkExactBatch2.exact_discriminants` (target `TASK-20261016-892515`) checks, per curve, $q+1-t=hn$, $t^2-4q=D_K f^2$, the product of the factors, the squarefree part and the D_K construction, $D_K \equiv 0,1 \pmod 4$ and the degree bound, reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).